

MIT WORLD PEACE UNIVERSITY



Health & Lifestyle Data for Diabetes Prediction

SUBJECT:

Regression Analysis

SUBMITTED BY

Shraddha G. Sonawane
M.Sc. Statistics

INSTITUTION

MIT World Peace University

ACADEMIC YEAR

2025 – 2026

-Under the guidance of Mr. Hemant Kulkarni

OBJECTIVE:

The main objective of this study is to analyse the impact of health and lifestyle factors on diabetes and to identify the most significant predictors using regression techniques. The study aims to develop a reliable model for predicting diabetes risk and to provide useful insights for early detection and prevention through improved lifestyle choices.

Data Description (Regressors & Target Variable)

The dataset contains multiple variables used for regression analysis. We have Secondary data with 21 col and 500 observations. Each column is described below along with its role:

Target Variable (Dependent Variable)

Diabetes Risk Score

Regressor Variables (Independent Variables)

1. **Age**
 - Type: Numerical
 - Description: Age of the individual in years
 - Role: Regressor
2. **Smoking Status**
 - Type: Categorical (e.g., Smoker / Non-smoker)
 - Description: Indicates whether the individual smokes
 - Role: Regressor
3. **Physical Activity (minutes per week)**
 - Type: Numerical
 - Description: Total minutes of physical activity performed weekly
 - Role: Regressor
4. **Diet Score**
 - Type: Numerical
 - Description: Score representing quality of diet (higher = healthier diet)
 - Role: Regressor
5. **Screen Time (hours per day)**
 - Type: Numerical
 - Description: Average daily screen exposure in hours

6. Family History of Diabetes

- Type: Binary (0 = No, 1 = Yes)
- Description: Indicates genetic predisposition to diabetes
- Role: Regressor

7. BMI (Body Mass Index)

- Type: Numerical
- Description: Measure of body fat based on height and weight
- Role: Regressor

8. Triglycerides

- Type: Numerical
- Description: Blood triglyceride level (indicator of metabolic health)
- Role: Regressor

```
## LOAD DATASET
```

```
data <- read_excel("C:/Users/User/Desktop/diabetes(500).xlsx")
```

```
head(data)
```

```
## MODEL
```

```
model <- lm(diabetes_risk_score ~ ., data = data)
```

```
model
```

```
summary(model)
```

OUTPUT:

Call:

```
lm(formula = diabetes_risk_score ~ ., data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.4907	-0.8117	-0.2054	0.7068	6.1578

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.4988216	1.1407595	2.190	0.0290 *
Age	0.2752443	0.0046613	59.048	< 2e-16 ***
Gender	0.0399287	0.1097162	0.364	0.7161

Education	-0.0642950	0.0679662	-0.946	0.3446
`Income level`	-0.0107976	0.0512252	-0.211	0.8331
`Smoking status`	-0.2862834	0.0709492	-4.035	6.36e-05 ***
alcohol_consumption_per_week	0.0177890	0.0398482	0.446	0.6555
physical_activity_minutes_per_week	-0.0357813	0.0007368	-48.561	< 2e-16 ***
diet_score	-0.5240171	0.0321152	-16.317	< 2e-16 ***
sleep_hours_per_day	0.0374307	0.0500299	0.748	0.4547
screen_time_hours_per_day	0.2961789	0.0221631	13.364	< 2e-16 ***
family_history_diabetes	16.2079716	0.1387315	116.830	< 2e-16 ***
hypertension_history	0.2469168	0.1308361	1.887	0.0597 .
bmi	0.5213721	0.0190658	27.346	< 2e-16 ***
systolic_bp	-0.0021720	0.0049371	-0.440	0.6602
diastolic_bp	0.0082064	0.0068446	1.199	0.2311
heart_rate	0.0073260	0.0069153	1.059	0.2900
cholesterol_total	0.0028476	0.0017635	1.615	0.1070
triglycerides	0.0130306	0.0013824	9.426	< 2e-16 ***
insulin_level	0.0057223	0.0116882	0.490	0.6247
hba1c	-0.0410048	0.0689087	-0.595	0.5521

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.198 on 478 degrees of freedom

Multiple R-squared: 0.9823, Adjusted R-squared: 0.9816

F-statistic: 1327 on 20 and 478 DF, p-value: < 2.2e-16

INTERPRETATION:

The model is highly significant ($p < 0.001$).

It explains 98.2% of the variation in diabetes risk score.

Family history of diabetes is the most important factor increasing risk.

Higher age, BMI, screen time, and triglycerides increase diabetes risk.

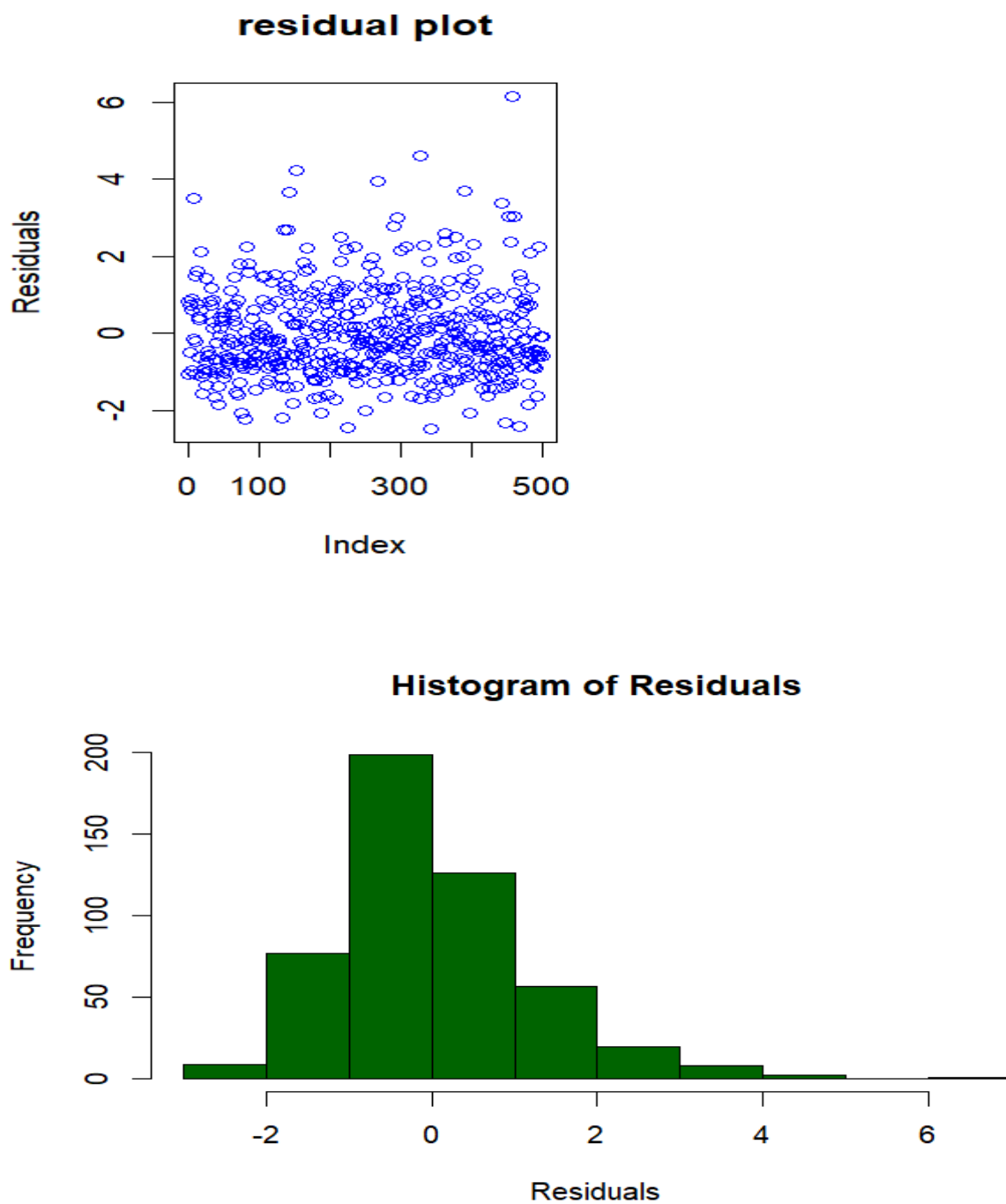
More physical activity and a better diet lower diabetes risk.

- **RESIDUAL ANALYSIS**

```
par(mfrow=c(1,2))
```

```
plot(model$residuals,main="residual plot",ylab="Residuals",col="blue")
```

```
hist(model$residuals,main="Histogram of  
Residuals",xlab="Residuals",col="darkgreen")
```



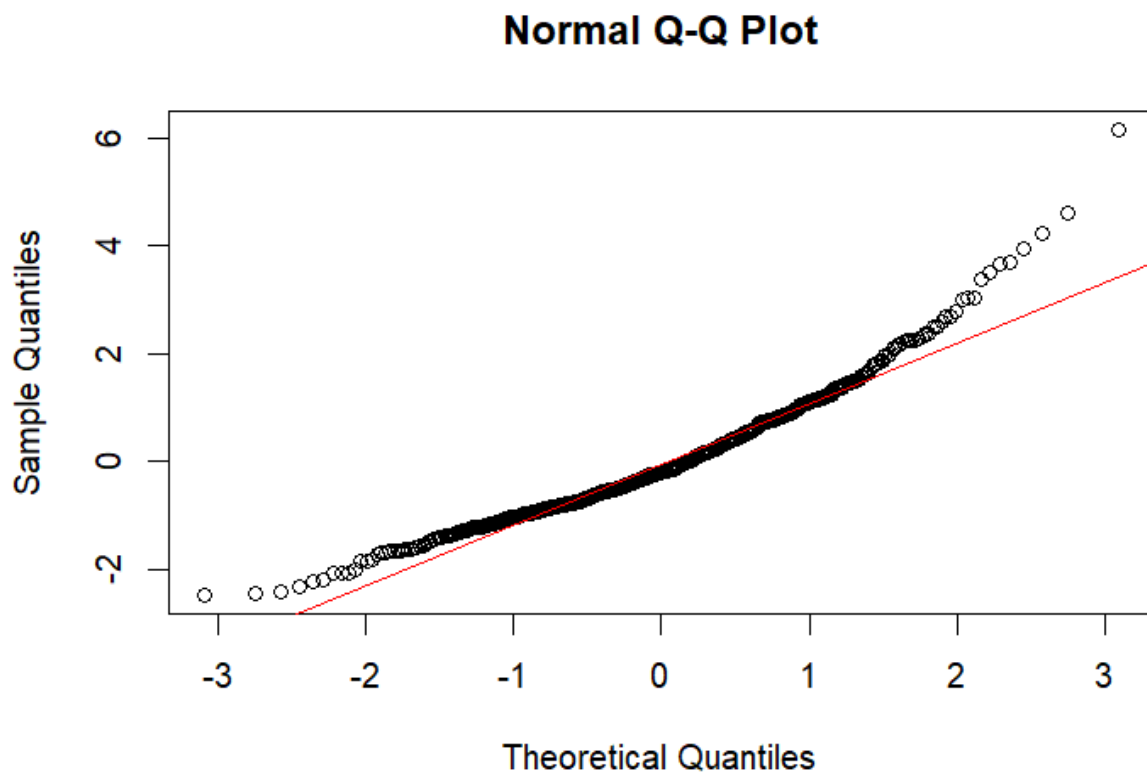
INTERPRETATION:

- Residuals are approximately normally distributed with slight right skewness.
- Residuals are centred around zero, showing good model fit.
- No clear pattern in residual plot, so linearity is satisfied.
- A few outliers are present, but overall, the model assumptions are reasonably met.

#qq plot

qqnorm(res)

qqline(res, col = "red")



```
ks.test(res, "pnorm", mean(res), sd(res))
```

output:

Asymptotic one-sample Kolmogorov-Smirnov test

data: res

D = 0.081931, p-value = 0.002464

alternative hypothesis: two-sided

Since the p-value is less than 0.05, the null hypothesis of normality is rejected. This indicates that the residuals are not normally distributed.

We Apply Transformation:

1)

```
ks.test(res, "pnorm", mean(re), sd(res))
```

Asymptotic one-sample Kolmogorov-Smirnov test

data: res

D = 0.15433, p-value = 9.507e-11

alternative hypothesis: two-sided

2) Box-Cox transformation:

```
ks.test(res, "pnorm", mean(res), sd(res))
```

Asymptotic one-sample Kolmogorov-Smirnov test

data: res

D = 0.077552, p-value = 0.001945

alternative hypothesis: two-sided

After applying both log and Box–Cox transformations, the residuals still did not follow a normal distribution perfectly. However, since the p-value remains below 0.05, residuals are still not perfectly normal.

- **TEST OF MULTICOLLINEARITY**

```
library(car)
```

```
vif (model)
```

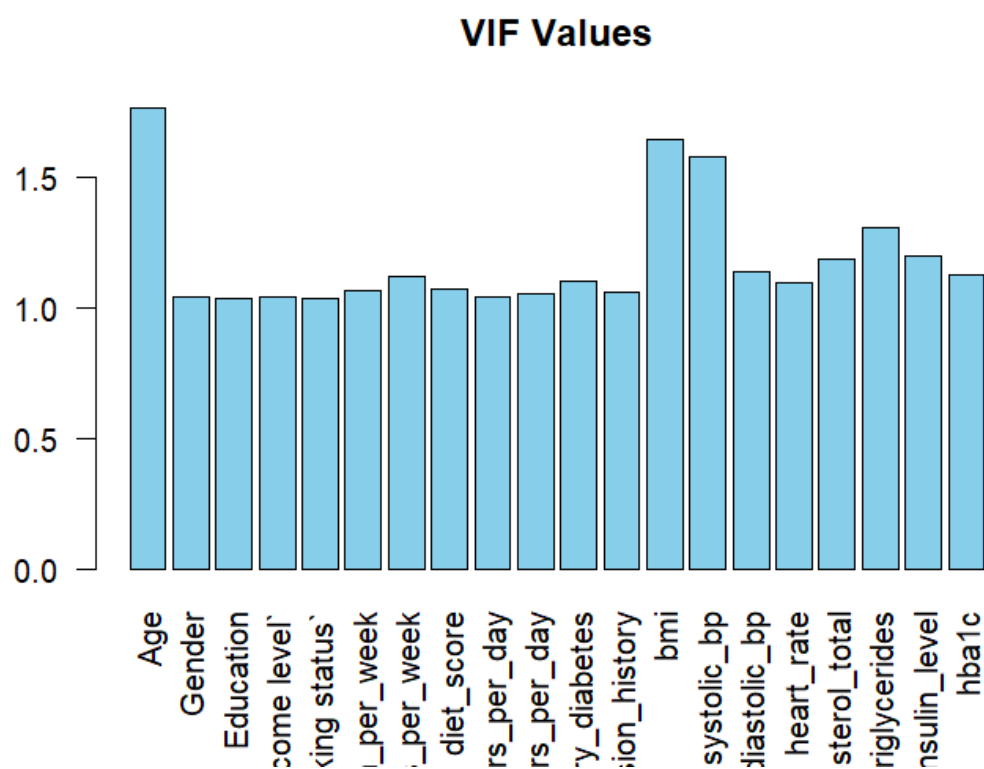
```
vif_values <- vif(model)
```

```
barplot(vif_values,
```

```
  main = "VIF Values",
```

```
  col = "skyblue",
```

```
  las = 2) # makes labels parallel
```



INTERPRETATION:

All VIF values are close to 1 and well below 5, indicating **no multicollinearity among the predictor variables.**

```
#conditional number
```

```
X <- model.matrix(model)[, -1] # remove intercept
```

```
X_scaled <- scale(X) # standardize variables
```

```
kappa(X_scaled)
```

OUTPUT:

```
2.973888
```

- **Conditional indices**

```
library(olsrr)
```

```
ci=ols_eigen_cindex(model)
```

```
ci
```

```
# Convert to data frame
```

```
ci_df= as.data.frame(ci)
```

```
# Check column names
```

```
colnames(ci_df)
```

```
# Plot (use correct column name)
```

```
plot(ci_df$`Condition Index`, type = "b", pch = 19,
```

```
  xlab = "Index",
```

```
  ylab = "Condition Index",
```

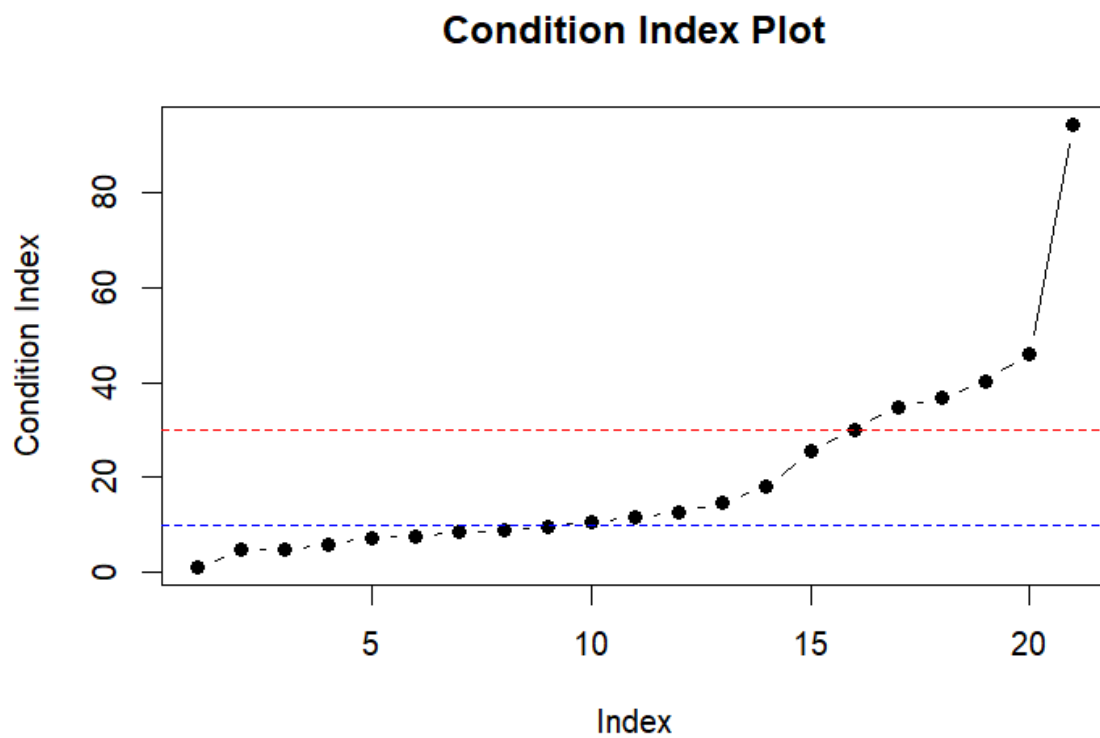
```
  main = "Condition Index Plot")
```

```
# Add reference lines
```

```
abline(h = 10, col = "blue", lty = 2)
```

```
abline(h = 30, col = "red", lty = 2)
```

PLOT:



OUTPUT:

The condition index plot shows values exceeding 30, indicating the presence of multicollinearity among a combination of Variables. Although individual VIF values are low, the condition index suggests dependency among groups of variables.

- **SUBSET SELECTION**

```
## MODEL
```

```
model = lm(diabetes_risk_score ~ ., data = data)
```

```
model
```

```
summary(model)
```

```
##subset selection
```

```
# Full model (start point)
```

```
full_model =lm(diabetes_risk_score ~ ., data = data)
```

```
full_model
```

```
# Stepwise selection (both directions: forward + backward)
```

```
step_model = step(full_model, direction = "both", trace =TRUE)
```

```
step_model
```

OUTPUT:

Start: AIC=200.99

diabetes_risk_score ~ Age + Gender + Education + `Income level` +
 `Smoking status` + alcohol_consumption_per_week + physical_activity_min
 utes_per_week +
 diet_score + sleep_hours_per_day + screen_time_hours_per_day +
 family_history_diabetes + hypertension_history + bmi + systolic_bp +
 diastolic_bp + heart_rate + cholesterol_total + triglycerides +
 insulin_level + hba1c

	Df	Sum of Sq	RSS	AIC
- `Income level`	1	0.1	686.3	199.04
- Gender	1	0.2	686.4	199.13
- systolic_bp	1	0.3	686.5	199.20
- alcohol_consumption_per_week	1	0.3	686.5	199.20
- insulin_level	1	0.3	686.6	199.24
- hba1c	1	0.5	686.8	199.36
- sleep_hours_per_day	1	0.8	687.0	199.58
- Education	1	1.3	687.5	199.93
- heart_rate	1	1.6	687.9	200.16
- diastolic_bp	1	2.1	688.3	200.49
<none>			686.2	200.99
- cholesterol_total	1	3.7	690.0	201.71

- hypertension_history	1	5.1	691.4	202.70
- `Smoking status`	1	23.4	709.6	215.71
- triglycerides	1	127.6	813.8	284.07
- screen_time_hours_per_day	1	256.4	942.6	357.40
- diet_score	1	382.2	1068.5	419.93
- bmi	1	1073.6	1759.8	668.92
- physical_activity_minutes_per_week	1	3385.5	4071.7	1087.51
- Age	1	5005.7	5692.0	1254.67
- family_history_diabetes	1	19595.5	20281.8	1888.73

Step: AIC=199.04

diabetes_risk_score ~ Age + Gender + Education + `Smoking status` +
 alcohol_consumption_per_week + physical_activity_minutes_per_week +
 diet_score + sleep_hours_per_day + screen_time_hours_per_day +
 family_history_diabetes + hypertension_history + bmi + systolic_bp +
 diastolic_bp + heart_rate + cholesterol_total + triglycerides +
 insulin_level + hba1c

	Df	Sum of Sq	RSS	AIC
- Gender	1	0.2	686.5	197.18
- alcohol_consumption_per_week	1	0.3	686.6	197.24
- systolic_bp	1	0.3	686.6	197.24
- insulin_level	1	0.3	686.6	197.28
- hba1c	1	0.5	686.8	197.41
- sleep_hours_per_day	1	0.8	687.1	197.62
- Education	1	1.3	687.6	197.97
- heart_rate	1	1.6	687.9	198.20
- diastolic_bp	1	2.1	688.4	198.54
<none>			686.3	199.04
- cholesterol_total	1	3.8	690.1	199.82
- hypertension_history	1	5.1	691.4	200.74
+ `Income level`	1	0.1	686.2	200.99
- `Smoking status`	1	23.5	709.8	213.84
- triglycerides	1	127.7	814.0	282.19
- screen_time_hours_per_day	1	256.5	942.8	355.51
- diet_score	1	386.3	1072.6	419.84
- bmi	1	1091.2	1777.5	671.90
- physical_activity_minutes_per_week	1	3389.4	4075.7	1086.00
- Age	1	5023.4	5709.7	1254.22
- family_history_diabetes	1	19601.4	20287.7	1886.88

Step: AIC=197.18

diabetes_risk_score ~ Age + Education + `Smoking status` + alcohol_consumption_per_week +
 physical_activity_minutes_per_week + diet_score + sleep_hours_per_day +
 screen_time_hours_per_day + family_history_diabetes + hypertension_history +
 bmi + systolic_bp + diastolic_bp + heart_rate + cholesterol_total +
 triglycerides + insulin_level + hba1c

	Df	Sum of Sq	RSS	AIC
- alcohol_consumption_per_week	1	0.3	686.8	195.36
- systolic_bp	1	0.3	686.8	195.37
- insulin_level	1	0.3	686.8	195.41
- hba1c	1	0.5	687.0	195.56
- sleep_hours_per_day	1	0.8	687.3	195.74
- Education	1	1.2	687.7	196.09
- heart_rate	1	1.6	688.1	196.32
- diastolic_bp	1	2.2	688.7	196.76
<none>			686.5	197.18
- cholesterol_total	1	3.8	690.3	197.95
- hypertension_history	1	5.0	691.5	198.83
+ Gender	1	0.2	686.3	199.04
+ `Income level`	1	0.1	686.4	199.13
- `Smoking status`	1	23.4	709.9	211.88
- triglycerides	1	127.8	814.3	280.38
- screen_time_hours_per_day	1	260.2	946.6	355.52
- diet_score	1	386.1	1072.6	417.85
- bmi	1	1098.0	1784.5	671.86
- physical_activity_minutes_per_week	1	3389.2	4075.7	1084.00
- Age	1	5023.3	5709.8	1252.23
- family_history_diabetes	1	19610.4	20296.9	1885.10

Step: AIC=195.36

diabetes_risk_score ~ Age + Education + `Smoking status` + physical_activity_minutes_per_week +
 diet_score + sleep_hours_per_day + screen_time_hours_per_day +
 family_history_diabetes + hypertension_history + bmi + systolic_bp +
 diastolic_bp + heart_rate + cholesterol_total + triglycerides +
 insulin_level + hba1c

	Df	Sum of Sq	RSS	AIC
- systolic_bp	1	0.2	687.0	193.53
- insulin_level	1	0.4	687.1	193.62
- hba1c	1	0.5	687.3	193.75

- sleep_hours_per_day	1	0.7	687.5	193.87
- Education	1	1.2	687.9	194.23
- heart_rate	1	1.6	688.3	194.51
- diastolic_bp	1	2.2	688.9	194.96
<none>			686.8	195.36
- cholesterol_total	1	3.7	690.5	196.08
- hypertension_history	1	5.0	691.7	196.96
+ alcohol_consumption_per_week	1	0.3	686.5	197.18
+ Gender	1	0.2	686.6	197.24
+ `Income level`	1	0.1	686.7	197.32
- `Smoking status`	1	23.3	710.1	210.04
- triglycerides	1	132.1	818.8	281.15
- screen_time_hours_per_day	1	262.3	949.1	354.79
- diet_score	1	386.5	1073.2	416.14
- bmi	1	1099.0	1785.8	670.23
- physical_activity_minutes_per_week	1	3402.2	4089.0	1083.62
- Age	1	5023.4	5710.2	1250.26
- family_history_diabetes	1	19624.5	20311.3	1883.46

Step: AIC=193.53

diabetes_risk_score ~ Age + Education + `Smoking status` + physical_activity_minutes_per_week +
 diet_score + sleep_hours_per_day + screen_time_hours_per_day +
 family_history_diabetes + hypertension_history + bmi + diastolic_bp +
 heart_rate + cholesterol_total + triglycerides + insulin_level +
 hba1c

	Df	Sum of Sq	RSS	AIC
- insulin_level	1	0.3	687.3	191.78
- hba1c	1	0.6	687.5	191.94
- sleep_hours_per_day	1	0.7	687.7	192.04
- Education	1	1.2	688.2	192.40
- heart_rate	1	1.7	688.7	192.76
- diastolic_bp	1	2.3	689.3	193.18
<none>			687.0	193.53
- cholesterol_total	1	3.7	690.7	194.19
- hypertension_history	1	4.9	691.9	195.08
+ systolic_bp	1	0.2	686.8	195.36
+ alcohol_consumption_per_week	1	0.2	686.8	195.37
+ Gender	1	0.1	686.8	195.42
+ `Income level`	1	0.1	686.9	195.49
- `Smoking status`	1	23.6	710.5	208.35
- triglycerides	1	132.7	819.6	279.64

- screen_time_hours_per_day	1	262.2	949.2	352.86
- diet_score	1	386.4	1073.4	414.21
- bmi	1	1138.3	1825.3	679.14
- physical_activity_minutes_per_week	1	3432.9	4119.9	1085.38
- Age	1	6678.7	7365.7	1375.30
- family_history_diabetes	1	19636.2	20323.2	1881.75

Step: AIC=191.78

diabetes_risk_score ~ Age + Education + `Smoking status` + physical_activity_minutes_per_week +

diet_score + sleep_hours_per_day + screen_time_hours_per_day + family_history_diabetes + hypertension_history + bmi + diastolic_bp + heart_rate + cholesterol_total + triglycerides + hba1c

	Df	Sum of Sq	RSS	AIC
- hba1c	1	0.5	687.8	190.16
- sleep_hours_per_day	1	0.7	688.0	190.29
- Education	1	1.2	688.6	190.68
- heart_rate	1	1.7	689.0	191.02
- diastolic_bp	1	2.3	689.6	191.46
<none>			687.3	191.78
- cholesterol_total	1	3.7	691.0	192.42
- hypertension_history	1	4.9	692.3	193.35
+ insulin_level	1	0.3	687.0	193.53
+ alcohol_consumption_per_week	1	0.3	687.1	193.59
+ systolic_bp	1	0.2	687.1	193.62
+ Gender	1	0.1	687.2	193.67
+ `Income level`	1	0.0	687.3	193.75
- `Smoking status`	1	23.6	710.9	206.60
- triglycerides	1	132.5	819.8	277.73
- screen_time_hours_per_day	1	262.4	949.8	351.15
- diet_score	1	390.0	1077.3	414.05
- bmi	1	1283.4	1970.8	715.41
- physical_activity_minutes_per_week	1	3439.3	4126.6	1084.19
- Age	1	6709.1	7396.4	1375.38
- family_history_diabetes	1	19654.2	20341.5	1880.20

Step: AIC=190.16

diabetes_risk_score ~ Age + Education + `Smoking status` + physical_activity_minutes_per_week +

diet_score + sleep_hours_per_day + screen_time_hours_per_day + family_history_diabetes + hypertension_history + bmi + diastolic_bp + heart_rate + cholesterol_total + triglycerides

	Df	Sum of Sq	RSS	AIC
- sleep_hours_per_day	1	0.7	688.5	188.66
- Education	1	1.3	689.1	189.06
- heart_rate	1	1.7	689.6	189.40
- diastolic_bp	1	2.5	690.3	189.97
<none>		687.8	190.16	
- cholesterol_total	1	3.7	691.5	190.80
+ hba1c	1	0.5	687.3	191.78
- hypertension_history	1	5.1	692.9	191.82
+ insulin_level	1	0.3	687.5	191.94
+ systolic_bp	1	0.3	687.6	191.97
+ alcohol_consumption_per_week	1	0.2	687.6	191.98
+ Gender	1	0.2	687.7	192.04
+ `Income level`	1	0.0	687.8	192.13
- `Smoking status`	1	23.7	711.5	205.04
- triglycerides	1	132.5	820.3	276.05
- screen_time_hours_per_day	1	262.6	950.4	349.51
- diet_score	1	389.5	1077.3	412.05
- bmi	1	1286.4	1974.3	714.30
- physical_activity_minutes_per_week	1	3442.8	4130.6	1082.68
- Age	1	6904.0	7591.9	1386.39
- family_history_diabetes	1	21014.1	21702.0	1910.50

Step: AIC=188.66

diabetes_risk_score ~ Age + Education + `Smoking status` + physical_activity_minutes_per_week +
 diet_score + screen_time_hours_per_day + family_history_diabetes +
 hypertension_history + bmi + diastolic_bp + heart_rate +
 cholesterol_total + triglycerides

	Df	Sum of Sq	RSS	AIC
- Education	1	1.4	690.0	187.69
- heart_rate	1	1.8	690.4	187.99
- diastolic_bp	1	2.4	691.0	188.43
<none>		688.5	188.66	
- cholesterol_total	1	3.5	692.0	189.18
+ sleep_hours_per_day	1	0.7	687.8	190.16
- hypertension_history	1	4.8	693.4	190.16
+ hba1c	1	0.5	688.0	190.29
+ insulin_level	1	0.3	688.2	190.44
+ systolic_bp	1	0.3	688.3	190.47
+ alcohol_consumption_per_week	1	0.2	688.4	190.53

+ Gender	1	0.1	688.4	190.56
+ `Income level`	1	0.0	688.5	190.64
- `Smoking status`	1	24.5	713.0	204.08
- triglycerides	1	132.5	821.1	274.50
- screen_time_hours_per_day	1	261.9	950.5	347.53
- diet_score	1	391.7	1080.2	411.38
- bmi	1	1285.8	1974.3	712.31
- physical_activity_minutes_per_week	1	3442.8	4131.4	1080.77
- Age	1	6970.7	7659.3	1388.80
- family_history_diabetes	1	21030.1	21718.7	1908.89

Step: AIC=187.69

diabetes_risk_score ~ Age + `Smoking status` + physical_activity_minutes_per_week +

diet_score + screen_time_hours_per_day + family_history_diabetes + hypertension_history + bmi + diastolic_bp + heart_rate + cholesterol_total + triglycerides

	Df	Sum of Sq	RSS	AIC
- heart_rate	1	1.9	691.9	187.08
- diastolic_bp	1	2.7	692.6	187.63
<none>			690.0	187.69
- cholesterol_total	1	3.5	693.4	188.18
+ Education	1	1.4	688.5	188.66
- hypertension_history	1	4.5	694.5	188.94
+ sleep_hours_per_day	1	0.9	689.1	189.06
+ hba1c	1	0.5	689.4	189.31
+ insulin_level	1	0.3	689.6	189.44
+ systolic_bp	1	0.3	689.7	189.50
+ alcohol_consumption_per_week	1	0.1	689.8	189.59
+ Gender	1	0.1	689.8	189.60
+ `Income level`	1	0.0	689.9	189.66
- `Smoking status`	1	24.1	714.1	202.84
- triglycerides	1	133.5	823.4	273.94
- screen_time_hours_per_day	1	263.3	953.2	346.97
- diet_score	1	390.4	1080.4	409.46
- bmi	1	1290.7	1980.7	711.91
- physical_activity_minutes_per_week	1	3453.5	4143.4	1080.22
- Age	1	6974.8	7664.7	1387.16
- family_history_diabetes	1	21047.5	21737.5	1907.32

Step: AIC=187.08

diabetes_risk_score ~ Age + `Smoking status` + physical_activity_minutes_per_week +
 diet_score + screen_time_hours_per_day + family_history_diabetes +
 hypertension_history + bmi + diastolic_bp + cholesterol_total +
 triglycerides

	Df	Sum of Sq	RSS	AIC
- diastolic_bp	1	2.6	694.5	186.95
<none>			691.9	187.08
- cholesterol_total	1	3.3	695.2	187.48
+ heart_rate	1	1.9	690.0	187.69
+ Education	1	1.5	690.4	187.99
- hypertension_history	1	4.4	696.3	188.22
+ sleep_hours_per_day	1	1.0	690.9	188.35
+ hba1c	1	0.5	691.4	188.72
+ systolic_bp	1	0.4	691.5	188.80
+ insulin_level	1	0.4	691.5	188.81
+ alcohol_consumption_per_week	1	0.1	691.8	188.99
+ Gender	1	0.1	691.8	189.02
+ `Income level`	1	0.0	691.9	189.07
- `Smoking status`	1	25.3	717.2	202.99
- triglycerides	1	132.2	824.1	272.33
- screen_time_hours_per_day	1	264.3	956.2	346.52
- diet_score	1	394.5	1086.4	410.24
- bmi	1	1294.9	1986.8	711.46
- physical_activity_minutes_per_week	1	3742.0	4433.9	1112.03
- Age	1	6978.9	7670.8	1385.55
- family_history_diabetes	1	21049.9	21741.8	1905.42

Step: AIC=186.95

diabetes_risk_score ~ Age + `Smoking status` + physical_activity_minutes_per_week +
 diet_score + screen_time_hours_per_day + family_history_diabetes +
 hypertension_history + bmi + cholesterol_total + triglycerides

	Df	Sum of Sq	RSS	AIC
<none>			694.5	186.95
+ diastolic_bp	1	2.6	691.9	187.08
- cholesterol_total	1	3.5	698.0	187.49
+ heart_rate	1	1.8	692.6	187.63
+ Education	1	1.8	692.7	187.68
- hypertension_history	1	4.2	698.7	187.99
+ sleep_hours_per_day	1	0.9	693.5	188.27

```

+ hba1c          1    0.7  693.8 188.45
+ systolic_bp    1    0.5  694.0 188.59
+ insulin_level  1    0.4  694.1 188.64
+ Gender         1    0.2  694.3 188.82
+ alcohol_consumption_per_week 1    0.1  694.3 188.85
+ `Income level` 1    0.0  694.5 188.93
- `Smoking status` 1   24.7  719.1 202.35
- triglycerides  1   131.2  825.7 271.31
- screen_time_hours_per_day    1   264.5  959.0 345.99
- diet_score      1   394.0 1088.4 409.16
- bmi             1  1355.3 2049.7 725.02
- physical_activity_minutes_per_week 1  3778.2 4472.7 1114.38
- Age             1  7382.9 8077.4 1409.32
- family_history_diabetes      1 21055.3 21749.8 1903.60
> step_model

```

Call:

```

lm(formula = diabetes_risk_score ~ Age + `Smoking status` + physical_activity_
_minutes_per_week +
  diet_score + screen_time_hours_per_day + family_history_diabetes +
  hypertension_history + bmi + cholesterol_total + triglycerides,
  data = data)

```

Coefficients:

(Intercept)	Age	`Smoking status`
3.26953	0.27512	-0.29127
physical_activity_minutes_per_week	diet_score	screen_ti
me_hours_per_day		
-0.03603	-0.52499	0.29589
family_history_diabetes	hypertension_history	
bmi		
16.18879	0.22322	0.52746
cholesterol_total	triglycerides	
0.00275	0.01297	

Final model summary

```
summary(step_model)
```

OUTPUT:

Call:

```

lm(formula = diabetes_risk_score ~ Age + `Smoking status` + physical_activity_
_minutes_per_week +

```

diet_score + screen_time_hours_per_day + family_history_diabetes +
hypertension_history + bmi + cholesterol_total + triglycerides,
data = data)

Residuals:

Min	1Q	Median	3Q	Max
-2.4728	-0.8286	-0.2372	0.7008	6.0698

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.2695326	0.5802977	5.634	2.98e-08 ***
Age	0.2751205	0.0038197	72.027	< 2e-16 ***
`Smoking status`	-0.2912704	0.0699805	-4.162	3.73e-05 ***
physical_activity_minutes_per_week	-0.0360256	0.0006992	-51.526	< 2e-16 **
diet_score	-0.5249854	0.0315533	-16.638	< 2e-16 ***
screen_time_hours_per_day	0.2958924	0.0217028	13.634	< 2e-16 ***
family_history_diabetes	16.1887913	0.1330925	121.636	< 2e-16 ***
hypertension_history	0.2232190	0.1292468	1.727	0.0848 .
bmi	0.5274648	0.0170924	30.860	< 2e-16 ***
cholesterol_total	0.0027503	0.0017435	1.578	0.1153
triglycerides	0.0129682	0.0013505	9.602	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.193 on 488 degrees of freedom

Multiple R-squared: 0.9821, Adjusted R-squared: 0.9817

F-statistic: 2676 on 10 and 488 DF, p-value: < 2.2e-16

INTERPRETATION:

The model fits the data very well and explains about **98% of diabetes risk variation**. The main factors affecting diabetes risk are **age, family history, BMI, physical activity, diet, screen time, and triglycerides**.

• LASSO REGRESSION

```
library(glmnet)

y = data$diabetes_risk_score

X = model.matrix(diabetes_risk_score ~ ., data = data)[, -1]

X_scaled = scale(X)

set.seed(123)

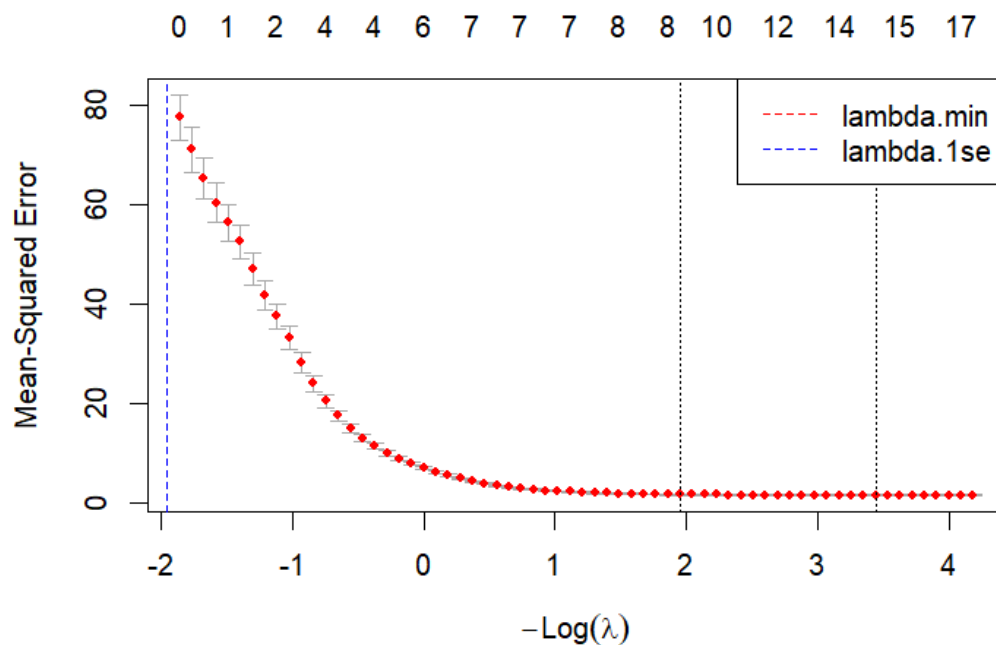
cv_lasso = cv.glmnet(X_scaled, y, alpha = 1) ##fit lasso

plot(cv_lasso) # cross validation curve

abline(v = log(cv_lasso$lambda.min), col = "red", lty = 2)

abline(v = log(cv_lasso$lambda.1se), col = "blue", lty = 2)

legend("topright",
      legend = c("lambda.min", "lambda.1se"),
      col = c("red", "blue"),
      lty = 2)
```



INTERPRETATION:

At the **left side (high λ)**:

Error is high in model and is too simple (many variables removed).

As λ decreases (moving right):

Error **drops quickly** and model improves because it includes more useful variables.

to find min and max lambda

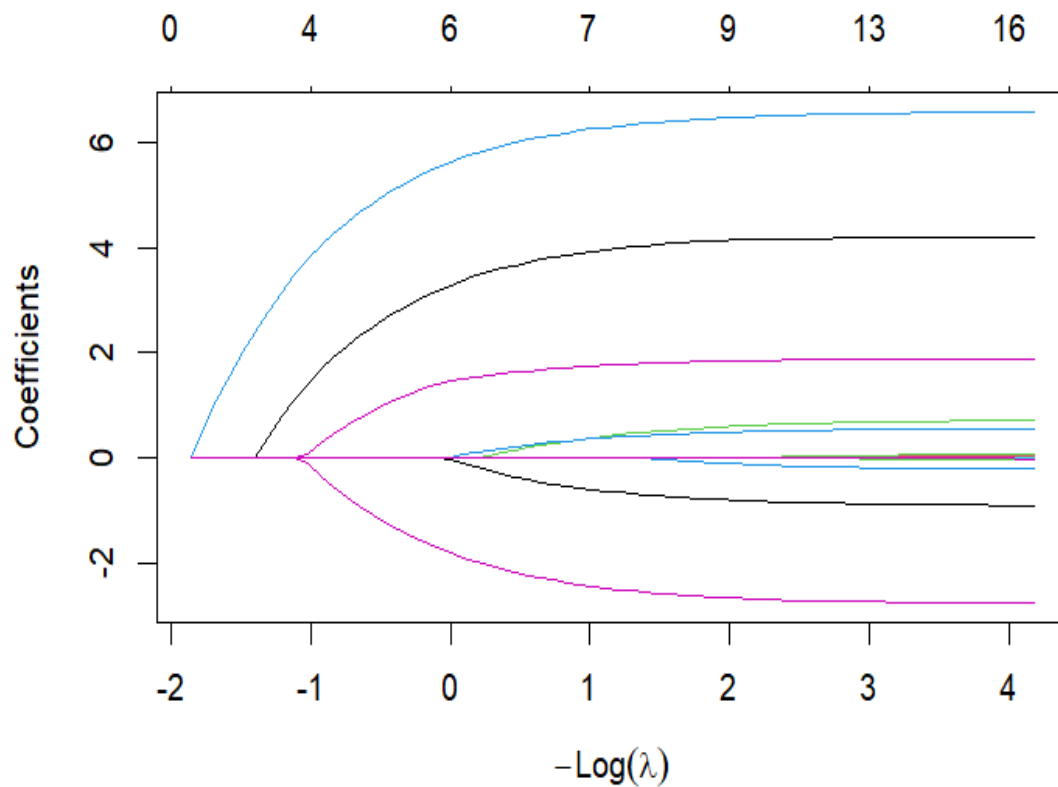
```
cv_lasso$lambda.min
```

```
0.03197675
```

```
cv_lasso$lambda.1se
```

```
0.1416769
```

```
plot(lasso_model, xvar = "lambda")
```



INTERPRETATION:

LASSO is selecting important variables by shrinking some coefficients to zero, and the graph shows how variables enter the model as the penalty decreases.

- Each line represents a different predictor (variable).
- The **y-axis** shows coefficient values (effect of each variable).
- The **x-axis** (**$-\log(\lambda)$**) shows how the penalty changes:
 - Left side strong penalty (λ large)
 - Right side weak penalty (λ small)

#subset selection

```
lasso_coef = coef(cv_lasso, s = "lambda.min")
```

```
selected_vars = rownames(lasso_coef)[lasso_coef[,1] != 0] #extract the selected variables
```

OUTPUT:

```
"(Intercept)"          "Age"
[3] "Education"           "`Smoking status`"
[5] "physical_activity_minutes_per_week" "diet_score"
[7] "sleep_hours_per_day"           "screen_time_hours_per_day"
[9] "family_history_diabetes"        "hypertension_history"
[11] "bmi"                        "diastolic_bp"
[13] "heart_rate"                 "cholesterol_total"
[15] "triglycerides"              "insulin_level"
```

```
selected_vars <- selected_vars[selected_vars != "(Intercept)"]
```

```
final_formula <- as.formula(
  paste("diabetes_risk_score ~", paste(selected_vars, collapse = "+"))
)
```

```
final_model <- lm(final_formula, data = data)
```

```
summary(final_model)
```

OUTPUT:

Call:

```
lm(formula = final_formula, data = data)
```

Residuals:

Min	1Q	Median	3Q	Max
-2.5154	-0.8265	-0.2116	0.6909	6.1985

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.0791560	0.9708203	2.142	0.0327 *
Age	0.2737435	0.0039377	69.519	< 2e-16 ***
Education	-0.0623056	0.0675456	-0.922	0.3568
`Smoking status`	-0.2873783	0.0704808	-4.077	5.33e-05 ***
physical_activity_minutes_per_week	-0.0357117	0.0007269	-49.129	< 2e-16 **
diet_score	-0.5226338	0.0317454	-16.463	< 2e-16 ***
sleep_hours_per_day	0.0345803	0.0495991	0.697	0.4860
screen_time_hours_per_day	0.2954538	0.0217615	13.577	< 2e-16 ***
family_history_diabetes	16.1835232	0.1333319	121.378	< 2e-16 ***
hypertension_history	0.2445781	0.1299317	1.882	0.0604 .
bmi	0.5198075	0.0183788	28.283	< 2e-16 ***
diastolic_bp	0.0089024	0.0067642	1.316	0.1888
heart_rate	0.0074573	0.0068594	1.087	0.2775
cholesterol_total	0.0028069	0.0017481	1.606	0.1090
triglycerides	0.0130703	0.0013539	9.654	< 2e-16 ***
insulin_level	0.0052956	0.0115764	0.457	0.6476

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.193 on 483 degrees of freedom

Multiple R-squared: 0.9823, Adjusted R-squared: 0.9817

F-statistic: 1784 on 15 and 483 DF, p-value: < 2.2e-16

CONCLUSION

1. The regression analysis shows that the model explains a **very high proportion of variability (~98% R^2)** in diabetes risk.
2. Key predictors identified are **family history, age, BMI, screen time, and triglyceride levels**, which positively influence diabetes risk.
3. **Lifestyle factors** such as physical activity and diet score have a **negative impact**, indicating that healthier habits reduce risk.
4. The model is statistically significant, confirming a strong relationship between predictors and the response variable.
5. Residual analysis suggests that **model assumptions are reasonably satisfied**, though minor deviations from normality exist.
6. Multicollinearity is not severe, indicating stable and reliable coefficient estimates

Comparison: Subset Selection (Stepwise) vs LASSO Regression

1. Method of Selection

- **Stepwise Selection**
 - Adds/removes variables step-by-step based on criteria like AIC
 - Uses forward, backward, or both directions
- **LASSO Regression**
 - Uses regularization (penalty λ)
 - Shrinks coefficients and automatically sets some to zero

2. Number of Variables Selected

- **Stepwise**
 - Selects **fewer variables**
 - Produces a compact model
- **LASSO**
 - Keeps **more variables**
 - Some coefficients are small but not exactly zero

3. Model Simplicity

- **Stepwise**
 - More **simple and interpretable**
 - Easy to explain in reports
- **LASSO**
 - Slightly **complex**
 - Includes additional variables

Both methods give **similar accuracy and model performance**, but their purpose differs. Stepwise is better suited for **clear interpretation and reporting**, while LASSO is preferred for **better prediction and handling complex data**.